Week 7 Cost-Benefit Memo

Evaluation of Complex Machine Learning Models for Emergency Severity Index Prediction

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1. Verdict

LightGBM is recommended as the preferred model for future deployment because it achieved the strongest overall balance of predictive performance, computational efficiency and practical interpretability among the models evaluated. While no model completely eliminates the challenges associated with clinical decision support, LightGBM provides the best combination of accuracy, speed and explainability for Emergency Severity Index (ESI) prediction.

2. Dataset and Methods Recap

The objective of this study was to determine whether a more advanced machine learning model could improve Emergency Severity Index (ESI) prediction compared with the baseline Logistic Regression model developed previously.

To ensure a fair comparison, every model was trained and evaluated using the same training and testing split from Week 6. A fixed random seed (`RANDOM_SEED = 42`) was used throughout the experiments to ensure reproducibility. Four models were evaluated: Logistic Regression, Random Forest, XGBoost and LightGBM.

Performance was assessed using accuracy, precision, recall, Macro F1-score, training time and inference time per prediction. Model interpretability was also considered because healthcare professionals must be able to understand how predictions are generated before they can confidently use a machine learning system in clinical practice.

3. Benchmark Comparison

Model	Accuracy	Precision	Recall	Macro F1	Training Time (s)	Inference Time (ms)	Interpretability
Logistic Regression	0.5924	0.4256	0.6436	0.4299	23.3209	11.5944	Excellent – coefficients directly explain predictions.
Random Forest	0.4906	0.3639	0.4338	0.3403	19.8585	91.0090	Moderate – feature importance provides global explanations.
XGBoost	0.6275	0.5539	0.5446	0.4964	26.6963	10.3501	Good – feature importance and SHAP values explain predictions.
LightGBM	**0.6619**	**0.5477**	**0.5157**	**0.4973**	**9.7358**	**10.0149**	Good – feature importance and SHAP values explain predictions.

The benchmark demonstrates that LightGBM produced the highest overall accuracy and Macro F1-score while also requiring the least training time and the fastest prediction time. Logistic Regression remained the easiest model to explain, but its predictive performance was lower than the leading gradient boosting models.

4. Arguments Supporting the Recommendation

The primary advantage of LightGBM is its strong predictive performance. It achieved the highest accuracy (66.19%) and Macro F1-score of all models evaluated, indicating a better overall ability to classify Emergency Severity Index categories than the baseline model or the other complex classifiers.

LightGBM also demonstrated excellent computational efficiency. Training completed in less than ten seconds, while individual predictions required only around 10 milliseconds. This makes the model suitable for emergency department environments where rapid decision support is essential.

Finally, although LightGBM is more complex than Logistic Regression, it still provides sufficient interpretability for practical use. Feature importance analysis offers a clear overview of the variables that influence predictions, while SHAP values can be used to explain individual patient predictions when required. This provides an appropriate balance between predictive performance and clinical transparency.

5. Arguments Against the Recommendation

Despite its advantages, LightGBM has several limitations. The model is less transparent than Logistic Regression because its predictions are generated through many interacting decision trees rather than a simple linear relationship. Additional explanation methods are therefore required when clinicians wish to understand individual predictions.

Implementation is also more technically demanding. Model tuning, deployment and ongoing monitoring require greater expertise than simpler statistical approaches, increasing maintenance requirements for healthcare organisations.

Finally, the current evaluation was performed using a single dataset. Further external validation is necessary to determine whether the model performs equally well across different hospitals, patient populations and clinical workflows.

6. Risks and Unknowns

Several important questions remain before LightGBM could be implemented in routine clinical practice. The model has not yet been evaluated prospectively within a live emergency department, meaning its real-world performance may differ from the results observed during testing.

In addition, further work should assess fairness across different patient groups, robustness to missing or incomplete data and long-term model stability. Successful deployment will also depend on clinician acceptance, regulatory approval and appropriate governance procedures. As a result, the model should be viewed as a decision-support tool that complements, rather than replaces, professional clinical judgement.

7. Recommendation

Based on the benchmark results, **LightGBM is recommended as the preferred model for future development and potential deployment**. It achieved the strongest overall balance between predictive performance, computational efficiency and practical interpretability, making it the most suitable model for supporting Emergency Severity Index prediction.

However, this recommendation does **not** solve every challenge associated with clinical decision support. LightGBM cannot eliminate the need for clinician oversight, guarantee fairness across all patient groups or ensure perfect prediction accuracy. Before deployment, the model should undergo external validation, fairness assessment and prospective clinical evaluation to confirm that it performs safely and reliably in routine healthcare settings.